ELEC4631 Tutorial 2

Do all of the following exercises by hand without using Matlab or any other computational aid.

1. Consider the quadratic form in $\mathbb{R}^3$

$$f(x) = 4x_1^2 + 6x_1x_2 + 4x_2^2 + 8x_2x_3 + 4x_3^2.$$

   (a) Write the quadratic form in the form $f(x) = x^T Ax$ for an appropriate real symmetric 3×3 matrix A.

   (b) Find a transformation V such that the new quadratic form $g(z) = f(Vz)$ in the variable z only contains sums of squares of the elements of z and no cross products.

   (c) Do the quadratic forms $f(x)$ and $g(z)$ have any definiteness properties?

2. For which values of $\epsilon > 0$ are the following two quadratic forms $f(x)$ and $g(x)$ on $\mathbb{R}^2$ positive definite and negative definite, respectively?

$$\begin{aligned} f(x) &= x_1^2 + 2\epsilon x_1x_2 + x_2^2, \\ g(x) &= -\epsilon x_1^2 - \epsilon x_1x_2 - (1-\epsilon)x_2^2. \end{aligned}$$

3. Compute the exponential of the following matrices:

   (a) $A = \begin{bmatrix} -9 & 8 \\ -12 & 11 \end{bmatrix}.$

   (b) $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$

   (c) $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0.5 & -0.5 \end{bmatrix}.$

4. Consider an LTI system with transfer function

$$H(s) = \frac{(s+2)(s+4)(s+6)}{(s+1)(s+3)(s+5)}$$

(a) Find the control canonical LTI state-space realisation of the transfer function and sketch the block diagram.

(b) Find the modal canonical LTI state-space realisation of the transfer function and sketch the block diagram.

(c) Find the observer canonical LTI state-space realisation of the transfer function and sketch the block diagram.

5. Consider an LTI system with input-output relation satisfying the linear second order ODE

$$y^{(2)}(t) + 2y^{(1)}(t) - 3y(t) = u^{(1)}(t) - u(t).$$

Answer the following questions:

(a) Write down the second order control canonical representation of this system.

(b) Solve the state-space equation of the control canonical representation for an arbitrary initial state x_0.

(c) Write down the second order observer canonical representation of this system.

(d) Solve the state-space equation of the observer canonical representation for an arbitrary initial state x_0.